

SIMATS ENGINEERING

ASSESSMENT TOOL I – Uncertainty Visualization Lab

Course: Data Handling and Visualization	Code: DSA06	CO: CO5
Duration: 180 min	Total Marks: 100	Reg No : 192324284 Name: V.P Sneha

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Uncertainty Visualization Lab

Weightage: 20%

Code of Conduct:

I V.P Sneha(192324284) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.P Sneha

Criteria	Understanding of Uncertainty 25%	Visualization of Uncertainty 25%	Application of Visualization Principles 20%	Organization 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Uncertainty Visualization Lab – Assessment Tool

R Programming Practical Questions with Datasets

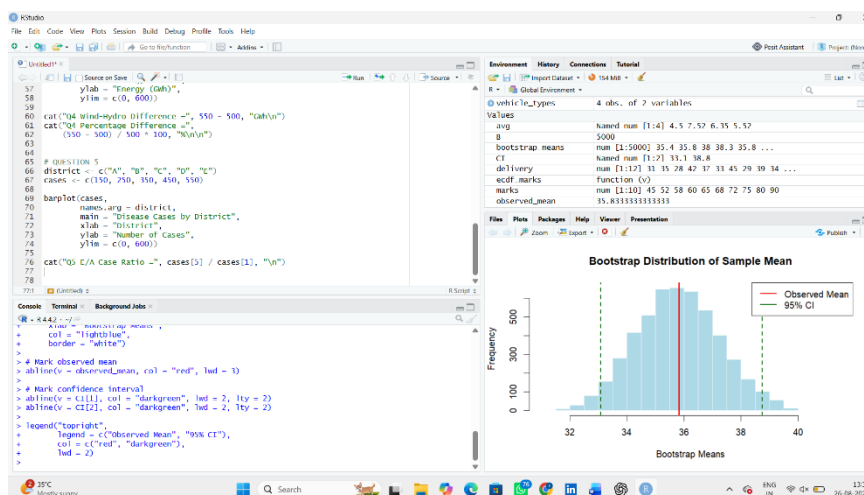
Instructions: Answer all five questions. Use R programming to construct the required visualization, interpret uncertainty, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Bootstrapping and Confidence Intervals – Point Estimate

A quality-control team records the following delivery times (in minutes) for 12 randomly selected orders. Using R, calculate the sample mean and obtain a 95% bootstrap confidence interval for the mean using at least 5,000 bootstrap resamples. Visualize the bootstrap distribution and mark the observed mean.

Data: 31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36

Expected R tasks: resampling, point estimate, confidence interval, histogram/density plot, interpretation.



Expected result

- Sample mean ≈ 35.83 minutes
- The bootstrap distribution will be centered around 35.83.
- The two green dashed lines represent the 95% bootstrap confidence interval.

Interpretation

The sample mean delivery time is approximately 35.83 minutes. The 95% bootstrap confidence interval gives a range of plausible values for the true average delivery time. The bootstrap distribution shows the sampling uncertainty around the observed mean.

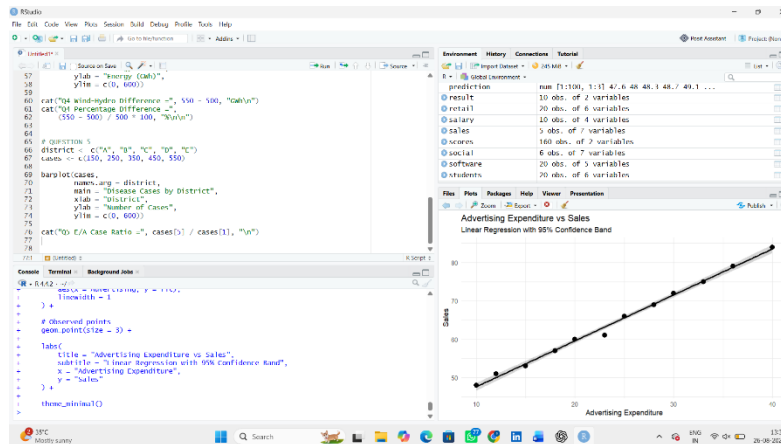
Q2. Uncertainty in Trend Estimation

The following data represent monthly advertising expenditure and sales. Fit a linear regression model in R to estimate the sales trend. Plot the observations with the fitted regression line and a 95% confidence band. Explain how the uncertainty changes across the range of advertising expenditure.

Data: Advertising = 10, 12, 15, 18, 20, 23, 25, 28, 30, 33, 36, 40;

Sales = 48, 51, 53, 57, 60, 61, 66, 69, 72, 75, 79, 84

Expected R tasks: `lm()`, prediction interval/confidence interval, `ggplot2` visualization, trend interpretation.



Interpretation

The regression line shows a positive relationship between advertising expenditure and sales. As advertising expenditure increases, sales also tend to increase.

The 95% confidence band is narrower near the center of the observed advertising range and becomes wider toward the lower and upper ends. This means the estimated mean sales are more uncertain at the extremes.

Q3. Overplotting, Jittering and Transparency

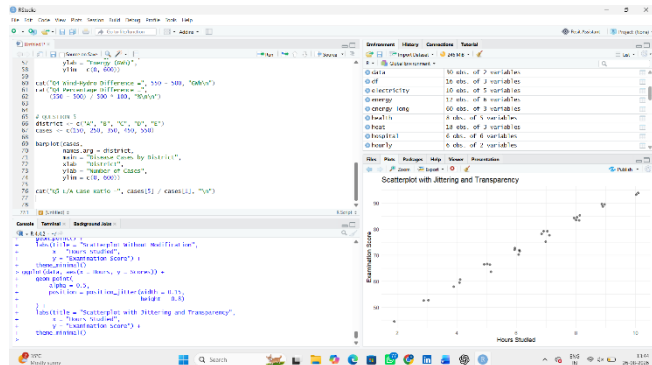
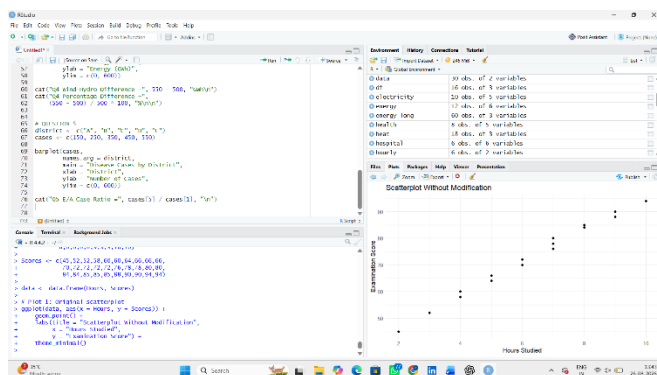
A survey records the number of hours studied and examination scores for 30 students. Several students have identical values. Create a scatterplot in R first without modification, then redesign it using partial transparency and jittering. Compare the two plots and explain how overplotting affects interpretation.

Data: Hours = 2,3,3,4,4,4,5,5,5,5,6,6,6,6,6,7,7,7,7,7,8,8,8,8,8,9,9,9,10,10;

Scores =

45,52,52,58,60,60,64,66,66,66,70,72,72,72,72,76,78,78,80,80,84,84,85,85,85,88,90,90,94,9

4 Expected R tasks: `geom_point()`, `alpha`, `position_jitter()`, comparison of overplotting.



Comparison

Original plot:

- Several observations overlap.
- It is difficult to see how many students have identical values.
- Some points may appear to be a single observation.

Jittered + transparent plot:

- Jitter slightly separates overlapping observations.
- Transparency makes areas with many observations darker.
- The distribution of students is easier to understand.

Interpretation

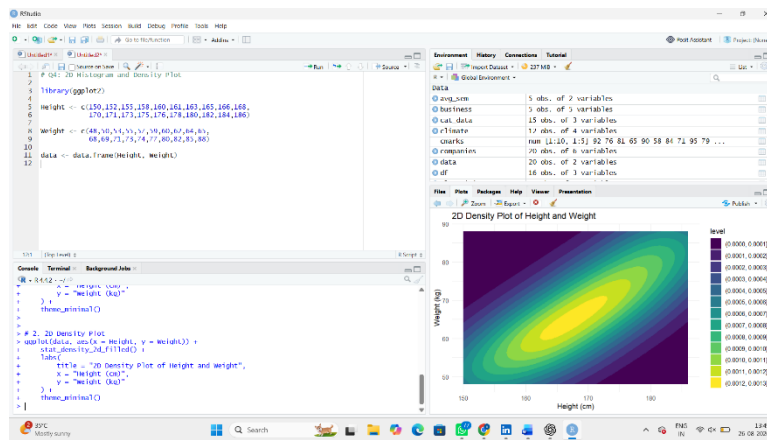
Overplotting can hide repeated observations and make the data appear less dense than it actually is. Jittering separates overlapping points, while transparency helps reveal areas with a high concentration of observations.

Q4. 2D Histogram and Density Estimation

A researcher measures height and weight for 20 participants. Construct a 2D histogram and a 2D density visualization in R. Use the plots to identify regions with high observation density. State which visualization communicates concentration more clearly and justify your choice.

*Data: Height = 150,152,155,158,160,161,163,165,166,168,170,171,173,175,176,178,180,182,184,186;
Weight = 48,50,53,55,57,59,60,62,64,65,68,69,71,73,74,77,80,82,85,88*

Expected R tasks: 2D binning, density estimation, contour/density layers, interpretation of concentration.



Interpretation

Both visualizations show that height and weight have a strong positive relationship. Higher values of height generally correspond to higher values of weight.

The 2D histogram divides the data into rectangular bins, while the 2D density plot provides a smoother representation of concentration.

The 2D density visualization communicates concentration more clearly because it shows the overall shape and high-density regions without depending heavily on arbitrary bin boundaries.

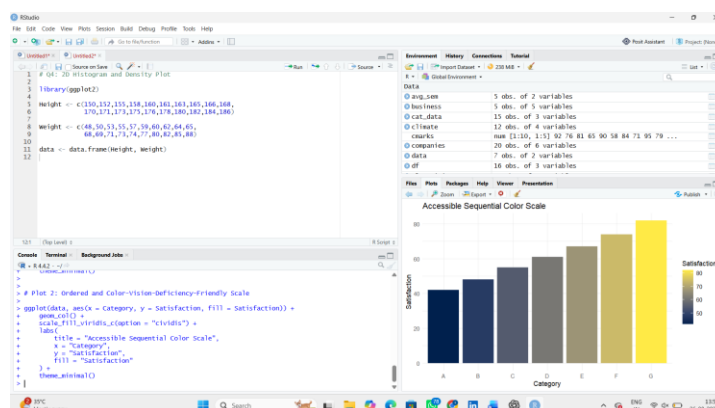
Q5. Color Scales, Accessibility and Misleading Visualization

The following category-wise satisfaction scores must be visualized in R. Create two plots: (1) a plot using a nonmonotonic/diverging color scheme that could mislead interpretation, and (2) a redesigned plot using an ordered, color-vision-deficiency-friendly sequential scale. Explain why the second design is more effective and identify the visualization pitfall in the first design.

Data: Category = A,B,C,D,E,F,G;

Satisfaction = 42,48,55,61,67,74,82

Expected R tasks: color mapping, scale selection, accessibility-aware redesign, critique of misleading color use.



Interpretation

The second visualization uses an ordered sequential color scale. The colors progress consistently with satisfaction values, making it easier to understand the increasing pattern.

The cividis palette is designed to be more accessible for people with color-vision deficiencies.

Main visualization pitfall

The first graph uses a diverging/nonmonotonic color scheme for data that have a single ordered direction. This can introduce a false visual meaning around the midpoint and exaggerate differences.

Suggested R packages: ggplot2, dplyr, boot (or base R).

Assessment focus: Correct R code, appropriate uncertainty visualization, interpretation, and identification of misleading design choices.